Liz Johnson

Job Objective: LLM/NLP/ML Engineer

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Education

Beihang University，BUAA - Master of Software Engineering (2021.09 - 2025.01)

- **Awards:** Beijing Outstanding Graduate, Beihang 2022 Outstanding Postgraduate
- **GPA:** 3.79 / 4.0
- **Key Courses:** Principles and Techniques of Artificial Intelligence, Fundamentals of Machine Learning, Deep Learning Techniques and Theory, Big Data, Industrial Robotics Theory and Experiments

Harbin Finance University，HFU - Bachelor of Accounting (2013.09 - 2017.06)

- **Awards:** National Scholarship, National Encouragement Scholarship, University Special Scholarship
- **GPA:** 4.15 / 5.0
- **Key Courses:** University Computer Programming (C Language), Fundamentals of Computer Science, Static Web Design, Graphics and Image Processing, E-commerce Overview, Accounting, Economics, Finance, Management

Project Experience

Enterprise Think Tank Q&A Platform - Beihang University Lab Project (2023.04 – Present)

- The platform developed a Q&A knowledge base for unstructured documents, such as enterprise policy files and industry research reports, to assist in official document drafting.
- Participated in building the RAG workflow and domain-specific model, involving fine-tuning and alignment training in three stages: Stage 1 used LLaMAFactory for SFT to address issues with model output format and style; Stage 2 applied DPO to align with human preferences; Stage 3 utilized Verl for GRPO to enhance the model’s data retrieval and aggregation capabilities.
- **Keywords:** SFT, DPO, GRPO, LLaMAFactory, Verl, LlamaIndex, RAG

Research Paper on LLM Inference Acceleration - Published in ICONIP2024(CCF-C) (2024.03 – 2024.11)

- Title: *Masker: Speculative Decoding for Long-Text QA with CopyHead*
- Summary: Proposed a novel inference acceleration method inspired by copy-paste operations. Integrated a CopyHead decoding module into the LLaMA source code to generate draft tokens (replicable segments), validated through pre-verification and parallel verification to confirm adoption.
- **Keywords:** LLM, LLaMA Source Code, Speculative Decoding

Research Paper on NLP-Based Log Anomaly Detection - Master's Thesis Project (2023.04 – 2024.12)

- Title: *Research on Log Anomaly Detection Based on Multi-Feature Fusion*
- Summary: Utilized heterogeneous graphs to fuse multiple log features and extracted richer semantic information via meta-paths. Designed a graph neural network model with a multi-head attention mechanism. Involved log parsing, grouping, representation, label estimation, heterogeneous graph construction, and GNN training.
- **Keywords:** NLP, PyTorch, PyTorch Geometric, Word Embedding, TF-IDF, HDBSCAN, HNSW, GNN

LizAgents - Personal Open-Source Project (2023.11 – 2024.02) github.com/liz-in-tech/LizAgents

- Implemented browser automation using Selenium and GPT-4V-Act.

- Integrated vision, memory, search, and document processing (CSV/PDF) capabilities, and experimented with agent decision-making, memory management, retrieval, and tool usage.
- **Keywords:** LangChain, Agent, RAG, Prompt Engineering, OpenAI API, Streamlit, Selenium, GPT-4V-Act

Work Experience

Whale Cloud Technology - Machine Learning Engineer (2022.03 – 2023.03)

- Worked on the TRaaS Technology Risk Prevention Platform
- Contributed to building anomaly detection models to identify potential risks and enable real-time alerts, and orchestrated atomic capabilities to generate emergency response plans.

NetEase - Backend Development Engineer (2021.07 – 2021.12)

- Worked on the CodeWave Low-Code Platform
- Participated in implementing bidirectional mapping from graphical interfaces to code generation: GUI <-> NetEase Application-Specific Language (NASL, JSON-formatted text) <-> Abstract Syntax Tree (AST) <-> Annotated Syntax Tree <-> Code Generation.

State Grid Corporation of China - Machine Learning Engineer (2018.08 – 2021.07)

- Worked on the Unified Payment Platform
- Contributed to building an abnormal transaction detection mechanism combining rule engines (blocking blacklists/IP anomalies, etc.), unsupervised learning (PCA + K-means for outlier detection), and supervised models (LightGBM integrating 200+ features).

Professional Skills

- **Programming Languages:** Python > Java > C/C++/JavaScript
- **LLM**: **Published papers, Experience modifying LLaMA source code, Training models (SFT/DPO/GRPO) on 16 A100 GPUs, Familiar with Verl distributed training framework and vLLM inference framework, understands distributed training principles.**
- **ML / NLP: PyTorch, PyTorch Geometric, Transformer, CNN/RNN/GNN, TF-IDF, Word Embedding**
- **Agent: LlamaIndex Contributions, LangChain Source Code, MCP**, Prompt Engineering, Ollama, Streamlit
- **RAG:** GraphRAG Source Code, RAG Workflow, RAG Evaluation Metrics, Neo4j, Vector Databases, HNSW Indexing
- **Backend:** MySQL, Redis, MQ, Microservices Architecture, Distributed Systems
- **DevOps:** Git, Linux, Docker
- **Blog:** https://liz-in-tech.github.io/blog
- **Certifications:** IBM Senior Software Engineer Certificate, National Software Talent Certificate, National Computer Rank Examination(NCRE) Level 2, IC3 (Internet Core Competency Certification) Digital Literacy certification, CET-6

Academic Competitions

- PAT (Zhejiang University Programming Ability Test, Java Language, 2019.03) **Score 87 points/ 100 points**
- MCM/ICM (Mathematical Contest in Modeling, 2016.04) **Honorable Mention**
- 7th Lanqiao Cup Competition (C Language, 2016.04) **Provincial First Prize**
- "HSBC Cup" SAS Data Analysis Competition (2015.10) **National Excellence Award**
- CUMCM (China Undergraduate Mathematical Contest in Modeling, 2015.09) **National Second Prize**
- 6th Lanqiao Cup Competition (C Language, 2015.04) **Provincial Second Prize**

张灵芝

求职意向: LLM/NLP/ML 算法工程师

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教育背景

北京航空航天大学 - 硕士 - 软件工程专业 (AI 方向) (2021.09 - 2025.01)

- **曾获奖项：** 北京市优秀毕业生，北航 2022 年优秀研究生
- **GPA：** 3.79 / 4.0
- **主修课程：** 人工智能原理与技术，机器学习基础，深度学习技术与理论，大数据，工业机器人理论与实验等

哈尔滨金融学院 - 本科 - 会计学专业 (2013.09 - 2017.06)

- **曾获奖项：** 国家奖学金，国家励志奖学金，校级特等奖学金
- **GPA：** 4.15 / 5.0
- **主修课程：** 大学计算机程序设计(C 语言)，大学计算机基础，静态网页设计，图形图像处理，电子商务概论，会计学，经济学，金融学，管理学

项目经验

企业智库问答平台 - 北航校内实验室项目 (2023.04 – 至今)

- 该平台对企业政策文件、行业研究报告等非结构化文档搭建问答知识库，辅助公文写作
- 参与构建RAG工作流和训练垂直领域模型，涉及三阶段的微调和对齐训练：第一阶段使用 LLaMAFactory 进行 SFT 解决模型输出格式和风格的问题，第二阶段进行 DPO 对齐人类偏好，第三阶段使用 Verl 进行 GRPO 强化提升模型找数据汇总数据的能力
- **关键词**：SFT，DPO，GRPO，LLaMAFactory，Verl，LlamaIndex，RAG

LLM 推测解码论文 - 小论文项目，ICONIP2024 (CCF-C) 接收 (2024.03 – 2024.11)

- 论文名称：《Masker: Speculative Decoding for Long-Text QA with CopyHead》
- 论文思路：该论文思路是模拟复制粘贴的操作来进行推理加速，在 LLaMA 源码的解码头处添加 CopyHead 解码头用于生成草稿 token 序列，即可被复制的部分，并通过预验证和并行验证来确认生成的草稿 token 序列是否可以被采纳
- **关键词**：LLM，LLaMA 源码，推测解码 Speculative Decoding

NLP 日志异常检测论文 - 毕业论文项目，被评为优秀论文 (2023.04 – 2024.12)

- 论文名称：《基于多特征融合的日志异常检测方法研究》
- 论文思路：该论文使用异质图将日志的多特征进行融合，并通过不同元路径提取更为丰富的语义信息，构建了基于多头注意力机制的图神经网络模型。涉及日志解析、日志分组、日志表示、标签估计、异质图的构造以及图神经网络的训练等步骤
- 关键词：NLP，PyTorch，PyTorch Geometric，Word Embedding，TF-IDF，HDBSCAN，HNSW，GNN

LizAgents - 个人开源项目 (2023.11 – 2024.02) github.com/liz-in-tech/LizAgents

- 基于 Selenium + GPT-4V-Act 实现浏览器自动化操作
- 集成 vision/memory/search/csv/pdf 等能力，对 Agent 的决策、记忆、检索、调用工具等进行了实践
- **关键词**：LangChain，Agent，RAG，Prompt Engineering，OpenAI API，Streamlit，Selenium，GPT-4V-Act

工作实习经历

浩鲸科技 - 机器学习工程师 - TRaaS技术风险防控平台 (2022.03 – 2023.03)

- 该平台旨在利用数据驱动的算法模型实现技术风险的提前预警与自动化响应，产品主要功能包括风险管理、故障诊断、应急预案、日常巡检、故障演练等
- 参与构建异常检测模型，实现对潜在风险的检测与实时预警，并对原子能力进行编排生成应急预案

网易 - 后端开发工程师 - 网易数帆CodeWave低代码平台 (2021.07 – 2021.12)

- 该平台通过图形化编程赋能非专业人员从事软件开发工作，是集研发测试部署运维为一体的一站式应用研发平台
- 参与实现从图形化界面到代码生成的双向映射：GUI <-> NetEase应用特定语言 NASL（JSON 格式化文本）<-> 抽象语法树 AST <-> 标注语法树 <-> 代码生成

国家电网 - 机器学习工程师 - 网上国网APP统一支付平台 (2018.08 – 2021.07)

- 该平台支持 27 省电力公司的统一下单支付、清分结算及跨省多户号批量缴费等业务
- 参与构建基于规则引擎(拦截黑名单/IP异常等基础策略)+无监督学习(基于 PCA + Kmeans 检测聚类离群点)+有监督模型(LightGBM 融合200+特征)的异常交易检测机制

专业技能

- **编程语言** : Python > Java > C/C++/JavaScript
- **LLM** : **有论文发表，改动 LLaMA 源码经验，16卡A100训练模型(SFT/DPO/GRPO)经验，verl 分布式训练框架，vLLM 推理框架，了解分布式训练原理**
- **ML / NLP** : **PyTorch，PyTorch Geometric，熟悉 Transformer / CNN / RNN / GNN / TF-IDF / Word Embedding 等技术**
- **Agent** : **LlamaIndex 开源项目贡献者，熟悉 LangChain 源码，熟悉 MCP**，Prompt 优化经验，Agent 开发经验，RAG 搭建经验，Ollama / Streamlit 经验
- **RAG** : 熟悉 GraphRAG 源码，RAG 工作流，RAG 评估指标，Neo4j 图数据库，向量数据库，HNSW 等索引算法
- **Backend:** MySQL, Redis, MQ, 微服务架构, 分布式系统
- **DevOps** : 熟悉 Git 版本控制，Linux 操作系统及其常用命令，Docker 容器化技术
- **博客** : https://liz-in-tech.github.io/blog
- **证书** : IBM 高级软件开发工程师证书 / 全国软件专业人才证书 / 国家计算机等级考试二级证书 / IC3 互联网和计算核心认证证书 / CET-6

学术竞赛

- PAT（浙江大学计算机编程能力考试，Java 语言参赛，2019.03） **总分87分/满分100分**
- MCM/ICM（美国大学生数学建模竞赛，2016.04） **国际二等奖（H 奖）**
- 第七届蓝桥杯大赛（C 语言参赛，2016.04） **省赛一等奖**
- 汇丰杯中国高校 SAS 数据分析大赛（2015.10） **国赛优秀奖**
- CUMCM（全国大学生数学建模竞赛，2015.09） **国赛二等奖**
- 第六届蓝桥杯大赛（C 语言参赛，2015.04） **省赛二等奖**